



Experiment – 9

AIM: Write the Queries to implement NESTED Queries.

Step-1: Write a Query to show table student.

```
mysql> SELECT * from student;
+-----+-----+-----+
| student_id | name   | course_id |
+-----+-----+-----+
| 1          | Alice  | 101       |
| 2          | Bob    | 102       |
| 3          | Charlie| 101       |
| 4          | David  | 103       |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

Step-2: Write a Query to show table course.

```
mysql> select * from course;
+-----+-----+
| course_id | course_name |
+-----+-----+
| 101       | Mathematics |
| 102       | Science     |
| 103       | History     |
+-----+-----+
3 rows in set (0.00 sec)
```

Step-3: Write a Query to find the names of the students who are enrolled in Mathematics.

```
mysql> SELECT name
-> FROM student
-> WHERE course_id = (SELECT course_id
-> FROM course
-> WHERE course_name = "Mathematics");
+-----+
| name   |
+-----+
| Alice  |
| Charlie|
+-----+
2 rows in set (0.06 sec)
```

Step-4: Write a Query to find the names of the students enrolled in courses that have only one student.



INSTITUTE OF ENGINEERING & TECHNOLOGY

MOHANLAL SUKHADIA UNIVERSITY, UDAIPUR

Name : Lavina Tamboli | Class: Btech-CSE IV sem) | Subject: DBMS Lab

```
mysql> SELECT name
-> FROM student s1
-> WHERE 1 = ( SELECT count(*)
-> FROM student s2
-> WHERE s1.course_id = s2.course_id);
+-----+
| name |
+-----+
| Bob   |
| David |
+-----+
2 rows in set (0.01 sec)
```

Step-5: Write a Query to find the names of the students who are enrolled in Science..

```
mysql> SELECT name
-> FROM student
-> WHERE course_id = (SELECT course_id
-> FROM course
-> WHERE course_name = "Science");
+-----+
| name |
+-----+
| Bob   |
+-----+
1 row in set (0.00 sec)
```

Step-6: Write a Query to find out names of the students who area either in Mathematics or in Science.

```
mysql> SELECT name
-> FROM student
-> WHERE course_id IN (SELECT course_id
-> FROM course
-> WHERE course_name IN ('Mathematics','Science'));
+-----+
| name |
+-----+
| Alice |
| Bob   |
| Charlie |
+-----+
3 rows in set (0.00 sec)
```

Step-7: Write a Query to find the names of the students who are not enrolled in History.



INSTITUTE OF ENGINEERING & TECHNOLOGY

MOHANLAL SUKHADIA UNIVERSITY, UDAIPUR

Name : Lavina Tamboli | Class: Btech-CSE IV sem) | Subject: DBMS Lab

```
mysql> SELECT name
-> FROM student
-> WHERE course_id NOT IN (SELECT course_id
-> FROM course
-> WHERE course_name = "History");
+-----+
| name  |
+-----+
| Alice |
| Bob   |
| Charlie |
+-----+
3 rows in set (0.00 sec)
```

Step-8: Write a Query to find the names of all the students only if Mathematics exist in the table course.

```
mysql> SELECT name from student
-> WHERE EXISTS (
-> SELECT * from course
-> WHERE course_name = "Mathematics"
-> );
+-----+
| name  |
+-----+
| Alice |
| Bob   |
| Charlie |
| David |
+-----+
4 rows in set (0.00 sec)
```

Step-9: Write a Query to find the names of those students whose course_id is greater than any course_id returned from Mathematics or History.

```
mysql> SELECT name from student
-> WHERE course_id > ANY (
-> SELECT course_id from course
-> WHERE course_name IN ('Mathematics','History')
-> );
+-----+
| name  |
+-----+
| Bob   |
| David |
+-----+
2 rows in set (0.01 sec)
```

Step-10: Write a Query to find the names of the students whose course_id is greater than all course_id from course table.

```
mysql> SELECT name from student
-> WHERE course_id > ALL (
-> select course_id FROM course
-> );
Empty set (0.00 sec)
```